

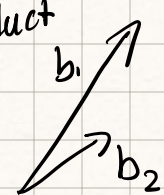
Thm (*) given any inner product in a fin. dim. vector space, there is some basis where the formula is the dot product

Lem: The matrix for the dot product is the non identity matrix iff we have a basis $[b_1, \dots, b_n]$ such that $\|b_i\|=1$ and $\langle b_i, b_j \rangle = 0 \ i \neq j$
Such a basis is called orthonormal

$$\underset{\text{//}}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = a_{11} = \|b_1\|^2$$

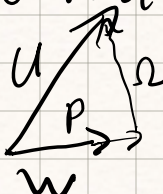
$\langle b_1, b_1 \rangle$ The fact that $\|b_1\|=1$ implies $\begin{bmatrix} 1 & ? \\ ? & 1 \end{bmatrix}$
the fact that $\langle b_i, b_j \rangle = 0$ implies that all other entries are zero
(*) is implied if we can find an orthonormal basis

Plan: 1) pick any basis
2) transform this basis into an orthonormal one

Ex: $\mathbb{R}^2_{\text{dot product}}$

 $\tilde{b}_1 := \frac{b_1}{\|b_1\|} \quad \tilde{b}_2 := \frac{b_2}{\|b_2\|}$
length 1 but not orthogonal

we need to decompose into $\perp$ and $\parallel$ parts

Given $u \in V$, we need to find a way to find the part of u that points in the same dir. as $w \in V$ and the part of u that is orthogonal to w



$$u = p + \omega \quad \langle w, u \rangle = \langle w, kw + \omega \rangle = k \langle w, w \rangle + \underbrace{\langle w, \omega \rangle}_0 = k \|w\|^2$$

$$K = \frac{\langle w, u \rangle}{\|w\|^2}$$

$$u = \underbrace{\frac{\langle u, w \rangle}{\|w\|^2} w}_{\text{this is the portion of } u \text{ in the dir of } w} + \Omega$$

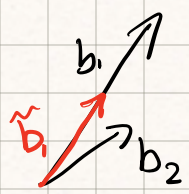
this is the portion of u in the dir of w

Def: the projection of u on w is: $\text{proj}_w u = \frac{\langle u, w \rangle}{\|w\|^2} w$

note that Ω can be calculated also

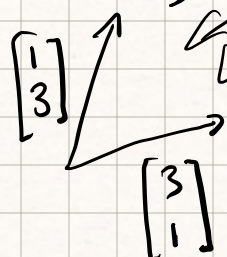
$$\Omega = u - \frac{\langle u, w \rangle}{\|w\|^2} w = u - \text{proj}_w u \quad \text{that is } u = p + \Omega, \langle p, \Omega \rangle = 0$$

back to ex in $\mathbb{R}^2$ w/ dot product

 $\tilde{b}_1 := \frac{b_1}{\|b_1\|}$ need the part of b_2 that is $\perp$ to b_1 , which we know is $(b_2 - \text{proj}_{b_1} b_2)$

$$\tilde{b}_2 := \frac{b_2 - \text{proj}_{b_1} b_2}{\|b_2 - \text{proj}_{b_1} b_2\|}$$

Ex: $\mathbb{R}^2$ dot product

 also our basis $\tilde{b}_1 = \frac{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}{\sqrt{\begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix}}} = \begin{bmatrix} 1/\sqrt{10} \\ 3/\sqrt{10} \end{bmatrix}$

$$\text{proj}_{b_1} b_2 = \frac{\langle \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \rangle}{\langle \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \rangle} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \frac{6}{10} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6/10 \\ 18/10 \end{bmatrix}$$

$$\tilde{b}_2 = \frac{\begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 6/10 \\ 18/10 \end{bmatrix}}{\|b_2 - \text{proj}_{b_1} b_2\|} = \frac{\begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix}}{\sqrt{\begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix} \cdot \begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix}}} = \frac{\begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix}}{\sqrt{\frac{640}{100}}} = \frac{\begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix}}{8\sqrt{1/10}}$$

$$= \begin{bmatrix} 3\sqrt{10}/10 \\ -\sqrt{10}/10 \end{bmatrix} \quad \langle \tilde{b}_1, \tilde{b}_2 \rangle = 3/10 - 3/10 = 0 \quad \checkmark$$